



Live App: arogyamesh.vercel.app Backend API: arogyamesh-api.onrender.com GitHub: [MKVISHNUVARDHAN/arogyamesh](https://github.com/MKVISHNUVARDHAN/arogyamesh)

1. Executive Summary & Impact

ArogyaMesh is an AI-driven, federated resilience autopilot designed specifically for India's **Primary Health Centre (PHC)** network. It replaces slow, error-prone manual registers with predictive digital twins, automated peer-to-peer redistribution, multi-district epidemic simulation, and privacy-preserving federated intelligence.

81

CONNECTED PHCS

9 Districts

AP · KA · TG TRI-BORDER

<10%

TARGET SHORTAGE RISK

2. The Last-Mile Public Health Problem

In rural Indian healthcare networks, medicine stockouts and facility overburdens cost lives. The current system suffers from three critical failure modes:

⚠️ Delayed Visibility

Stock arrives or depletes unrecorded. Manual tallying creates 24–72 hour lag, making epidemic response always reactive.

❌ Cannibalizing Transfers

Emergency medicine requests rob neighbouring PHCs of buffer stock without safety calculations, causing secondary stockouts.

🔒 Inter-State Data Silos

State health departments (Andhra, Karnataka, Telangana) cannot pool raw patient/facility data due to governance and residency mandates.

3. The ArogyaMesh 5-Pillar Architecture

1. Verifiable TruthScore & Digital Twins

Combines consumption patterns, attendance-to-bed capacity curves, and reporting lag into an objective 0–100 Data TruthScore. Filters phantom inventory from decision pipelines.

2. Guaranteed Safe Redistribution (MILP)

Greedy distance-optimal algorithm with post-transfer safety guarantees. A donor PHC cannot donate if its projected stockout risk exceeds 10%. Enforces First-Expired-First-Out (FEFO).

3. Multi-District Emergency Sandbox

Simulates epidemic surges (Dengue, Heatwave, Respiratory) with capacity overflows spilling over to nearest facilities, calculating unmet units before and after intervention.

4. Inter-State Federated Learning

Federated Averaging (FedAvg) trains demand models locally inside state boundaries. **Zero raw health data transfers** across borders; only model weights and sample counts are shared.

5. Offline-First PWA & Bilingual AI Copilot

Works seamlessly in rural low-bandwidth areas via IndexedDB client transaction queues and service workers. Includes **Arogya Copilot** powered by Groq (LLM) delivering sub-second, clinically grounded explanations in **English and Telugu script**.

1. Deployed Production Cloud Topology

Layer	Provider	Specification	Role & Responsibility
Frontend	Vercel	Next.js 16 (App Router), React 19, TypeScript, PWA	Command Center UI, Maps, Digital Twin visualizer, offline PWA
Backend API	Render	FastAPI, Uvicorn, Python 3.12 (Ohio, US East)	Optimization solver, simulation sandbox, federated aggregator
Database	Neon PostgreSQL	Serverless Postgres 17, PgBouncer pooler (AWS Ohio)	Persistent storage: 81 PHCs, 1,620 batches, 145,000 history points
AI Copilot	Groq Cloud	openai/gpt-oss-20b inference engine	Sub-second explanations in English & Telugu from structured evidence

2. Algorithmic Innovations in services/ml

Safe Redistribution Engine

Rule: Every proposed donor must pass a post-transfer safety check.

Computes required safety stock: $S = \mu \times H + z \times \sigma \times \sqrt{H}$.

Donors are evaluated using same-district first, then cross-district based on geographical Haversine distance. If post-donation risk exceeds 10%, the donor is rejected.

Zero Secondary Stockouts

TruthScore Calculation

Rule: Distinguish physical stock from verifiable usable stock.

Evaluates reporting lag (hours since last sync), batch expiration windows, and footfall anomalies. Generates an objective score:

$TruthScore = 100 - (LagPenalty + ExpiryPenalty + VariancePenalty)$.

Audit-Trail Verified

3. The Multi-District Epidemic Simulation

During an outbreak, overwhelmed PHCs overflow patients to nearest neighbour facilities within road transit range. ArogyaMesh calculates epidemic curves over a 14-day window:



4. Privacy-Preserving Federated Intelligence

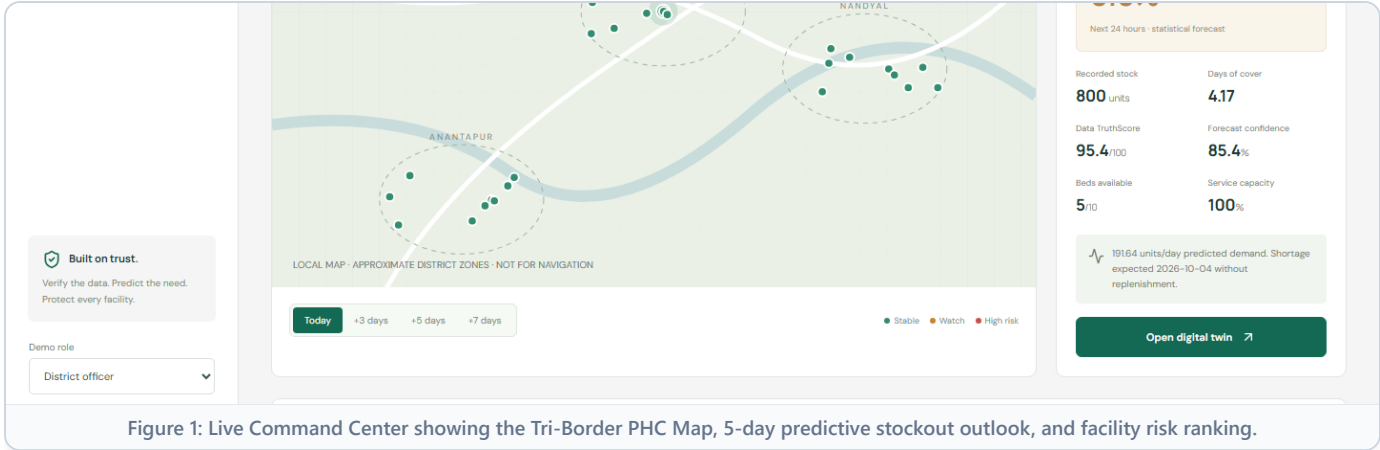
- 1

State Siled Workers: Andhra Pradesh (AP), Karnataka (KA), and Telangana (TG) workers access only their respective state database partitions.
- 2

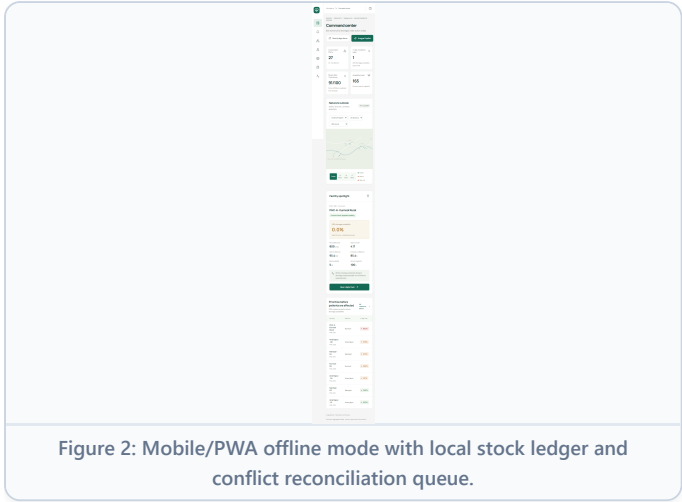
Local Model Training: Each state computes gradient updates on local ORS demand history (holding out final 18 days for strict validation).
- 3

Zero PII FedAvg Aggregator: The central server receives only weights vector $[w_0, w_1, w_2]$ and sample counts. Aggregated global weights reduce system-wide MAE without raw row transfers.

1. District Officer Command Center



2. Rural Operator Offline Station



3. Offline-First Resilience

⚡ Works Without Internet

Rural PHC health workers often operate in blackouts or zero-cellular zones. ArogyaMesh caches the application shell via **Service Worker** and stores dispensing events locally in **IndexedDB**.

1. Offline Dispense:

Updates local count instantly.

2. Transaction Queue:

UUID events held safely.

3. Reconnection:

Idempotent batch sync to Neon DB.

Tested & Verified under Browser Offline Mode

4. Arogya Copilot (Bilingual AI Supply Chain Assistant)

Native Groq Integration:

The AI Copilot translates complex optimization math, risk forecasts, and truth score audit trails into human-readable recommendations for district officers and medical officers.

"గ్రామీణ కర్షులు ప్రాథమిక ఆరోగ్య కేంద్రంలో ఓఆర్ఎస్ స్టాక్ రాబోయే 5 రోజుల్లో 92% ప్రమాదంలో ఉంది..."

(Answers generated in sub-second latency in both English and native Telugu script).

1. The 3-Minute Judge Demonstration Script

- 1

Clean Demo Reset: Open arogyamesh.vercel.app, ensure role is *District Officer*, and click **"Start judge demo"**. Resets reproducible synthetic state (seed 42).
- 2

Command Center & Predictive Slider: PHC-A currently has 800 ORS units and looks healthy. Toggle the **" +5 days "** slider: shortage risk spikes past 90% as demand outpaces stock.
- 3

Digital Twin Inspection: Click **"Open digital twin"**: verify attendance-to-bed capacity curves, consumption trends, and the 0–100 Data TruthScore ledger.
- 4

Find Safe Redistribution: Click **"Find safe redistribution"**. Algorithm identifies donors. It safely splits allocations between PHC-B and cross-district PHC-C without violating either donor's safety floor.
- 5

Approval & Dispatch: Click **"Approve plan"** → **"Simulate dispatch"**. Observe transactional inventory debits and FEFO batch deductions.
- 6

Emergency Simulator: Go to *Emergency Simulator*, select *Dengue surge* (250% demand, 14 days), and click **"Optimize response"** to show dynamic overflow routing and failure mitigation.
- 7

Federated Learning Round: In *Federated Intelligence*, click **"Train federated round"** to show AP, KA, TG training locally and aggregating weights with zero raw patient rows shared.
- 8

Offline Operation: Switch to *PHC operations*, click **"Go offline (demo)"**, record transactions, and click **"Reconnect"** to observe instantaneous queue reconciliation.

2. Verification & Quality Gates Summary

Verification Test Suite	Tools & Target	Result
Backend Integration & Models	pytest (14 test cases in tests/)	14 / 14 Passed (100%)
Code Formatting & Linting	Ruff (PEP 8, type rules, import order)	All checks passed
Frontend Type Safety	tsc --noEmit (TypeScript 5.9)	0 Errors
Offline Queue Unit Tests	Node test runner (tests/queue.test.mjs)	3 / 3 Passed
Database Scale & Migration	Neon Serverless Postgres 17	81 PHCs · 1,620 Batches · Seeded
End-to-End Stack Pre-check	scripts/demo-check.py	4 / 4 Live Services OK

ArogyaMesh: Ready for Hackathon Presentation

Live URLs: <https://arogyamesh.vercel.app> · <https://arogyamesh-api-6o84.onrender.com>